VVDEX FORGE

CERTIFIED EVALUATION ENVIRONMENTS

Robot catch temporal and tracking annotation

EVALUATION REPORT

2026-09-07

Evaluation scope:

meta/muse-spark-1.3

Prepared by:

VVDex Forge — generated directly from sealed run evidence

fr-20260907-023b5cdb · vvdex.annotation.robot-video-1 v0.2.0 · Independent model evaluation report

About this report

About VVDex Forge

VVDex Forge builds and certifies evaluation environments for AI agents. Every exam is sealed before any model sees it: the reference solution must pass, an empty submission must fail, the grader must reject near-misses, and the hidden tests never enter the agent's tree. Reports and the public catalog live at **vvdexops.com**.

Report integrity

This report is generated from run evidence; its digests verify authenticity. The sole test of a copy is the three-line procedure in Verification — a copy whose digests do not reproduce is not this report.

Scope

One exam · 10 rollout(s) · 24 tool steps and 2280s wall clock per rollout, stated in full in Evaluation configuration.

Boundary

Not a leaderboard. Not a general capability score. A post-seal benchmark: this run stamped no certification gate and rewrote nothing — not the exam, not the reference solution, not the grader, not the certified evidence.

Contents

01 Annotation evidence

Modality: video. **Rubric:** 1.1.

A robot-catching video is reviewed for temporal events, frame-level object boxes, track identity and occlusion.

Submission dd120ba7-a5d4-42fd-8c4a-7c312d18a602

Score	Items	Valid / invalid	Types
0.7638	6	6 / 0	box, classification, event, segment

Measured metrics

annotationCount	6	boxF1	1	boxIoU	0.579
boxPrecision	1	boxRecall	1	classificationAccuracy	1
classificationF1	1	classificationPrecision	1	classificationRecall	1
eventF1	1	eventPrecision	1	eventRecall	1
eventTemporalIoU	0.7125	matchedCount	6	mismatchCount	4
segmentF1	1	segmentPrecision	1	segmentRecall	1

segmentTemporalIoU	1	Track consistency on matched boxes	1

Track consistency depends on successful box matching. A zero can reflect inaccurate boxes even when track IDs are correct.

Validation error codes

None recorded

Disagreement

Agreement review	Not measured

Submission cc1b55f2-6d54-436d-8737-aa81aae3a220

Score	Items	Valid / invalid	Types
0.7003	6	6 / 0	box, classification, event, segment

Measured metrics

annotationCount	6	boxF1	0.5	boxIoU	0.5528
boxPrecision	0.5	boxRecall	0.5	classificationAccuracy	1
classificationF1	1	classificationPrecision	1	classificationRecall	1
eventF1	1	eventPrecision	1	eventRecall	1
eventTemporalIoU	0.5982	matchedCount	6	mismatchCount	5
segmentF1	1	segmentPrecision	1	segmentRecall	1
segmentTemporalIoU	0.9	Track consistency on matched boxes	1		

Track consistency depends on successful box matching. A zero can reflect inaccurate boxes even when track IDs are correct.

Validation error codes

None recorded

Disagreement

Agreement review	Not measured

Submission 415869af-1240-4a10-bf95-d12fccfcb96c

Score	Items	Valid / invalid	Types
0.7802	6	6 / 0	box, classification, event, segment

Measured metrics

annotationCount	6	boxF1	0	boxIoU	0.4656
boxPrecision	0	boxRecall	0	classificationAccuracy	1
classificationF1	1	classificationPrecision	1	classificationRecall	1
eventF1	1	eventPrecision	1	eventRecall	1
eventTemporalIoU	0.875	matchedCount	6	mismatchCount	3
segmentF1	1	segmentPrecision	1	segmentRecall	1
segmentTemporalIoU	1	Track consistency on matched boxes	1		

Track consistency depends on successful box matching. A zero can reflect inaccurate boxes even when track IDs are correct.

Validation error codes

None recorded

Disagreement

Agreement review	Not measured

Submission 8fe2d8d3-b887-4b2f-ae75-a28eda0bb8be

Score	Items	Valid / invalid	Types
0.6654	6	6 / 0	box, classification, event, segment

Measured metrics

annotationCount	6	boxF1	1	boxIoU	0.5797
boxPrecision	1	boxRecall	1	classificationAccuracy	1
classificationF1	1	classificationPrecision	1	classificationRecall	1

eventF1	0.5	eventPrecision	0.5	eventRecall	0.5
eventTemporalIoU	0.4667	matchedCount	6	mismatchCount	5
segmentF1	1	segmentPrecision	1	segmentRecall	1
segmentTemporalIoU	0.9	Track consistency on matched boxes	1		

Track consistency depends on successful box matching. A zero can reflect inaccurate boxes even when track IDs are correct.

Validation error codes

None recorded

Disagreement

Agreement review	Not measured

Submission 0dbb95df-4a7f-438d-b3c3-74a1e68017eb

Score	Items	Valid / invalid	Types
0.7908	6	6 / 0	box, classification, event, segment

Measured metrics

annotationCount	6	boxF1	0.5	boxIoU	0.4975
boxPrecision	0.5	boxRecall	0.5	classificationAccuracy	1
classificationF1	1	classificationPrecision	1	classificationRecall	1
eventF1	1	eventPrecision	1	eventRecall	1
eventTemporalIoU	0.875	matchedCount	6	mismatchCount	3
segmentF1	1	segmentPrecision	1	segmentRecall	1
segmentTemporalIoU	1	Track consistency on matched boxes	1		

Track consistency depends on successful box matching. A zero can reflect inaccurate boxes even when track IDs are correct.

Validation error codes

None recorded

Disagreement

Agreement review	Not measured

Submission 078d6f61-7dd8-4c2a-8f74-eb4049a5b391

Score	Items	Valid / invalid	Types
0.8217	6	6 / 0	box, classification, event, segment

Measured metrics

annotationCount	6	boxF1	1	boxIoU	0.5902
boxPrecision	1	boxRecall	1	classificationAccuracy	1
classificationF1	1	classificationPrecision	1	classificationRecall	1
eventF1	1	eventPrecision	1	eventRecall	1
eventTemporalIoU	0.875	matchedCount	6	mismatchCount	3
segmentF1	1	segmentPrecision	1	segmentRecall	1
segmentTemporalIoU	1	Track consistency on matched boxes	1		

Track consistency depends on successful box matching. A zero can reflect inaccurate boxes even when track IDs are correct.

Validation error codes

None recorded

Disagreement

Agreement review	Not measured

Submission 2efda1c5-b89e-4014-a4cf-161e5c35599f

Score	Items	Valid / invalid	Types
0.6826	6	6 / 0	box, classification, event, segment

Measured metrics

annotationCount	6	boxF1	0	boxIoU	0.459
boxPrecision	0	boxRecall	0	classificationAccuracy	1

classificationF1	1	classificationPrecision	1	classificationRecall	1
eventF1	1	eventPrecision	1	eventRecall	1
eventTemporalIoU	0.6125	matchedCount	6	mismatchCount	5
segmentF1	1	segmentPrecision	1	segmentRecall	1
segmentTemporalIoU	0.9524	Track consistency on matched boxes	1		

Track consistency depends on successful box matching. A zero can reflect inaccurate boxes even when track IDs are correct.

Validation error codes

None recorded

Disagreement

Agreement review	Not measured

Submission be8a7b18-8bd0-4ebd-9588-0211ca846bc1

Score	Items	Valid / invalid	Types
0.7473	6	6 / 0	box, classification, event, segment

Measured metrics

annotationCount	6	boxF1	1	boxIoU	0.5295
boxPrecision	1	boxRecall	1	classificationAccuracy	1
classificationF1	1	classificationPrecision	1	classificationRecall	1
eventF1	1	eventPrecision	1	eventRecall	1
eventTemporalIoU	0.7125	matchedCount	6	mismatchCount	4
segmentF1	1	segmentPrecision	1	segmentRecall	1
segmentTemporalIoU	1	Track consistency on matched boxes	1		

Track consistency depends on successful box matching. A zero can reflect inaccurate boxes even when track IDs are correct.

Validation error codes

None recorded

Disagreement

Agreement review	Not measured

Measured grader aggregates only. Missing metrics or disagreement are not measured, never zero. Reference answers and item-level labels are excluded. Certification and the evidence digest are recorded in this report's verification sections.

02 Executive summary

MODELS TESTED

1

CERTIFIED PASSES

3

of 8 graded rollouts

SUBMITTED

8 of 8

graded rollouts that called submit

NOT GRADED — LANE ERRORS

2

of 10 attempts

Purpose

This report presents a controlled evaluation of 1 model lane(s) against the certified exam vv `dex.annotation.robot-video-1` (v0.2.0, family annotation). Every lane received the identical frozen workspace, tool contract and limits; grading ran afterwards, in isolation, against hidden tests no lane ever saw. The purpose is to state, with reproducible evidence, what each lane did and where it stopped.

Outcome

3 of 8 graded rollouts passed the independent hidden tests and were certified — 2 of 10 attempts not graded (lane errors)

3 of 8 graded rollouts passed. meta/muse-spark-1.3 completed the applicable annotation workflow and were accepted by the independent annotation grader. The most common way to fail was `submitted_failed` — submitted annotations rejected by the independent grader.

Key findings

F-01	meta/muse-spark-1.3 passed the independent hidden tests
F-02	submitted_failed × 5 (meta/muse-spark-1.3)
F-03	model_api_failure × 2 (meta/muse-spark-1.3)

Each finding is stated in full, with evidence, in the Findings section; recommendations for acting on them follow it. Elapsed wall clock for the whole engagement: 2824.7s.

03 Exam definition

In scope. The ability of each configured model lane to complete this one certified task end to end — inspect the asset, annotate against the declared rubric, validate, submit, and be accepted by the independent annotation grader — within 24 tool steps and 2280s of wall clock, at 10 rollout(s) per lane. Behavioural measurements along the way (tool discipline, grounding, overclaim) are in scope because they are read from the recorded trace, not judged by another model.

Out of scope. General model quality, performance on other task families, prompt-tuning on the lane's behalf, and any comparison this run's sample size cannot support. A lane's API failure is reported as infrastructure, never as model capability.

The instrument. Exam `vvdex.annotation.robot-video-1` was certified before this run: its reference solution passes, an empty submission fails, its grader distinguishes near-misses, and its sealed evidence is unchanged by this evaluation (see Certification evidence).

What was measured

EXAM	VERSION	FAMILY
vvdex.annotation.robot-video-1	0.2.0	annotation

ROLLOUTS
10 (10 per model)

Where the defect came from

This exam has no upstream repository to credit: the defect, the reference solution and the hidden suite are VVDex's own.

04 Certification evidence

Why this exam is trustworthy. Every pillar below was established before any model saw the exam, loaded from the sealed evidence matching this run's exam fingerprint, and unchanged by this evaluation.

FROZEN BASELINE

FAIL

expected — an empty submission over 2 run(s) must not pass

→

GOLD / REFERENCE

PASS

expected — the reference solution over 2 run(s) must pass

GRADER CONTROLS

19 / 19

The grader accepted the reference and rejected every near-miss.

ATTACK PROBES

9 / 9 blocked

0 trivial exploit(s) found.

SEALED EVIDENCE

verified

Sealed 2026-09-06 over 13 artifact(s).

RUNTIME

sha256:679b36225a1f83238efba8c71b9cde3c24caaeacc9b682ae92819cb55eec328f

network policy: deny

CERTIFICATION BUNDLE DIGEST

`0a0a51360cde69737c9b5caef895b8b7c2a5e2d66da89617d8776ac724b019b9`

EXAM FINGERPRINT

`85e026007af17db692614084…`

Full value in Appendix B; the contract digest this run was executed against.

05 Model outcomes

Denominators. Attempts requested: 10. Graded (evaluable) rollouts: 8. Not graded — lane errors: 2. A lane error is a rollout the API or the runtime ended before the hidden grader ran: it is listed, excluded from every rate and pass@k, and never silently dropped. Passes are certified passes: hidden grader exit 0 with integrity verified.

MODEL	ATTEMPTS	GRADED	PASSED	MODEL FAILS	LANE ERRORS	SUBMITTED	MEAN TIME	TYPICAL DEEPEST STAGE
meta/muse-spark-1.3 meta/muse-spark-1.3	10	8	3 / 8	5	2	8 / 8	267.1s	Graded

Every rollout, with its own deepest stage and grade, is in Appendix A.

Success rate, confidence and pass@k

Every rate on this page is a measurement of a finite number of rollouts, so every rate carries the interval that measurement actually supports. n is the evaluable count — attempts minus rollouts the lane ended before grading; those are listed under "not graded" and sit in no rate, interval or pass@k.

MODEL	ATTEMPTS	N	NOT GRADED	PASSES	RATE	95% INTERVAL	SEEDS	PASS@1	PASS@3	PASS@5	PASS@8
meta/muse-spark-1.3	10	8	2	3	37.5%	[14%, 69%]	0–9	37.5%	82.1%	98.2%	100.0%

Interval and pass@k methods are stated in full in Appendix D. pass@k is shown for k = 1, 3, 5, 8; the sealed record carries every k up to n.

Model comparison

One model ran in this record, so there is nothing to compare it with. A comparison table across models appears here when a run covers more than one.

06 Stage funnel

Recorded annotation actions and independent grader verdicts. Missing capabilities are N/A; each stage is measured independently.

STAGE	OPENROUTER:META/MUSE-SPARK-1.3
Inspect	8 / 8
Annotate	8 / 8
Validate	8 / 8
Submit	8 / 8
Graded	8 / 8
Passed	3 / 8

Deepest stage reached

MODEL	DEEPEST (ANY ATTEMPT)	TYPICAL (MOST ATTEMPTS)	ANNOTATION QA PASSES	NOT GRADED
openrouter:meta/muse-spark-1.3	Passed	Graded	3 / 8	2

The matrix counts evaluable annotation attempts; lane errors are shown separately.
Successful recorded actions establish each stage independently.

07 Findings

Every finding below is derived from the recorded data of this run — none is an opinion.
Severity: PASS a lane succeeded · ATTENTION a capability gap · LANE / INFRA infrastructure, not capability · NOTE a property of the measurement.

F-01 PASS

meta/muse-spark-1.3 passed the independent hidden tests

Severity: PASS	Evidence: Model outcomes · Stage funnel · Appendix A

3 of 8 graded rollouts completed the applicable annotation workflow and were accepted by the independent annotation grader. This demonstrates the task is solvable under the published limits.

F-01 · derived from the recorded run data

F-02 ATTENTION

submitted_failed × 5 (meta/muse-spark-1.3)

Severity: ATTENTION	Evidence: Appendix A · Appendix C

Submitted annotations rejected by the independent grader.

F-02 · derived from the recorded run data

F-03 LANE / INFRA

model_api_failure × 2 (meta/muse-spark-1.3)

Severity: LANE / INFRA	Evidence: Appendix A · Appendix C

The model endpoint failed to answer. Not a verdict about the model's ability. These rollouts are excluded from any capability reading.

F-03 · derived from the recorded run data

08 Recommendations

One recommendation per observed failure mode, addressed to the lane it affects. Each traces back to a finding above and to the raw trace in Appendix A.

R-01 PROBLEM

Submitted annotations rejected by the independent grader.

AFFECTED LANES

meta/muse-spark-1.3

WHAT WE WOULD LOOK AT FIRST

Review the measured annotation metrics and validator error codes against the declared rubric.

R-02 PROBLEM

The model endpoint failed to answer. Not a verdict about the model's ability.

AFFECTED LANES

meta/muse-spark-1.3

WHAT WE WOULD LOOK AT FIRST

The lane failed, not the model; these rollouts carry no signal about capability and are labeled accordingly.

09 Verification

RUN ID

live-20260907T082233Z

EXAM FINGERPRINT

85e026007af17db692614084f12ff40c5bbf2348bcfda0a01deb95aad8928f4c

The contract digest this run was executed against.

Tamper-evidence

Three digests, and how to check each one. If any of them does not reproduce, this page is not the page that was generated for this run.

THIS REPORT

17ab141810dc6c72a69cab783660cab0224b52f699e60052bc17239b2ddab857

sha256 over the report's UTF-8 bytes with every occurrence of the report digest itself replaced by 64 '0' characters

EVAL RECORD

079d54cc556ec585dc806c66514f025eca04c9325bf64854e9421669b3f09056

sha256 of fr-20260907-023b5cdb.json as written beside this file.

RUN EVIDENCE BUNDLE

023b5cdb0bec7cdb5402199e372974bbbaacea8cfea639dc9fbf118bbcef9d1b

The digest the run's own bundle computed over itself.

CERTIFIED EVIDENCE SEAL

0a0a51360cde69737c9b5caef895b8b7c2a5e2d66da89617d8776ac724b019b9

The exam's sealed evidence, established before this run.

Measurement history

This is the first recorded run on this exam. There is nothing to compare it against yet, and a single measurement is not a trend.

Verifying it, in three lines

1. `vvdex-env verify-report --report fr-20260907-023b5cdb.html`
2. `or by hand: replace every occurrence of the report digest above with 64 zeros, then sha256 the file — it must equal that digest`
3. `shasum -a 256 fr-20260907-023b5cdb.json must equal the eval-record digest, and the bundle digest must equal bundleSha256 in the run's bundle file`

The full artifact-by-artifact digest table, and the reproduction protocol, are in Appendix B.

This is the executive report. The full verifiable evidence report for the same run — every rollout, artifact digests, redacted transcripts and methods — is `fr-20260907-023b5cdb.pdf`, generated from exactly the same sealed record.

Attestation

This report was generated end to end by VVDex Forge from the sealed evidence of the evaluation run described above. Measurement values were populated from recorded artifacts, not manually entered. The digests in Verification let any holder of this file confirm that the report, the eval record and the run evidence are the ones this run produced — without contacting VVDex.

VVDex Forge · vvdexops.com	**2026-09-07**	**vvdex-env 0.1.0**
Issuer — certified evaluation environments	Date of issue	Engine · verifiers pin c521e4146026

No upstream repository: this exam is VVDex’s own.

Post-seal benchmark. It does not stamp any certifying gate and did not rewrite the exam, the gold solution, the grader or the certified evidence.

— END OF REPORT · fr-20260907-023b5cdb —